



ASIA PACIFIC UNIVERSITY TECHNOLOGY & INNOVATION

EE008-3-3-MSES MICROPROCESSOR SYSTEMS AND EMBEDDED SOFTWARE

TITLE	BALANACE REHABILITATION MONITORING AND CONTROL SYSTEM
NAME & STUDENT ID	YAP PENG KUN (TP084179)
INTAKE	APD3F2601ME
LECTURER	IR. TS. DR. DENESH A/L SOORIAMOORTHY
DATE	10 th September 2026

Table of Contents

1.0	Balance Rehabilitation Monitoring and Control Subsystem (Yap Peng Kun TP084179)	5
1.1	Input/Output Interface Design and Program Development	6
1.1.1	Subsystem Overview	6
1.1.2	Hardware and Electronic Interface Design	7
1.1.3	Program Development and Balance Monitoring Logic	16
1.1.4	Web-Based User Interface and Data Management	23
1.1.5	Design Considerations	33
1.2	Enhancement / Innovation of the Subsystem.....	36
1.2.1	ASI-Based Balance Assessment	36
1.2.2	Web-Based Real-Time Monitoring and Control	37
1.2.3	Firebase-Based Patient and Session Storage.....	37
1.2.4	Multi-Session Rehabilitation Trend Assessment	38
1.2.5	Comparison with Existing Rehabilitation Systems and Product Justification.	38
1.3	Results and Discussion	41
1.3.1	Subsystem Testing and Results	41
1.3.2	Problems Encountered and Troubleshooting	46
1.3.3	Limitations	49
1.3.4	Future Improvements	50
1.4	Conclusion	52
1.5	Reference	54

List of Figure

Figure 1.1: Overall hardware setup of the Balance Rehabilitation Platform.	8
Figure 1.2: A 38-pin ESP32 connected in an expansion board.....	8
Figure 1.3: Hardware architecture of the Balance Rehabilitation subsystem.	10

Figure 1.4: Arrangement of the four 50kg half-bridge load cells below the platform.....	11
Figure 1.5: two HX711 ADC modules used in the platform.	12
Figure 1.6: The 0.96-inch OLED display secured in a 3D-printed holder.....	14
Figure 1.7: A DFPlayer Mini module used to amplify audio tracks to the speaker.	15
Figure 1.8: A 4 Ω 3W speaker for playing real-time auditory feedback.	15
Figure 1.9: The software architecture of the balance rehabilitation subsystem.....	17
Figure 1.10: The web interface will display the ESP32 connection status.	24
Figure 1.11: Real-time weight distribution of the patient is displayed.	25
Figure 1.12: Evaluation metrics calculated during the active session.	26
Figure 1.13: Session control and timer in the web interface.....	27
Figure 1.14: One of the session data stored in the Firebase for PATIENT009.	29
Figure 1.15: Patient has to be selected before starting a session or acquiring session history.	29
Figure 1.16: Average ASI graph generated from accessing PATIENT009.	30
Figure 1.17: Balanced time and performance score graphs generated from PATIENT009's progression.....	31
Figure 1.18: Progress assessment generated by the system based on past performance of PATIENT009.....	32
Figure 1.19: All the patient data are stored and saved to Firebase Realtime Database.	43
Figure 1.20: Detailed graphical analysis based on previous session history for PATIENT009.	44
Figure 1.21: Balance time and performance score graphs based on PATIENT009's historical data.....	44
Figure 1.22: Trend assessment generated by the system based on the patient's past performance.	45
Figure 1.23: Numerical history of the patient's past rehabilitation sessions.	45
Figure 1.24: Efficient architecture for Firebase integration.....	48

List of Table

Table 1.1: Summary of all input and output in the system.	7
Table 1.2: Main electronic interfaces of the balance rehabilitation subsystem.	10
Table 1.3: Audio feedback logic with respect to the patient's weight distribution.	22
Table 1.4: HTTP endpoints and their respective functions.	23
Table 1.5: Three levels of the system architecture.....	24

Table 1.6: Comparison between proposed system with existing products in the market.	39
Table 1.7: Load cell calibration test results and relative error.	41
Table 1.8: Result obtained for weight distribution test.	42
Table 1.9: Summary of problems encountered and their solutions.	48

List of Equation

Eq. 1.1	12
Eq. 1.2	13
Eq. 1.3	18
Eq. 1.4	18
Eq. 1.5	19
Eq. 1.6	19
Eq. 1.7	20
Eq. 1.8	21
Eq. 1.9	21
Eq. 1.10	21
Eq. 1.11	41

1.0 Balance Rehabilitation Monitoring and Control Subsystem (Yap Peng Kun TP084179)

The Balance Rehabilitation Monitoring and Control Subsystem was developed as one of the individual subsystems of the overall Rehabilitation Monitoring System. The subsystem aims to support balance rehabilitation by measuring the patient's body weight distribution between the left and right sides of the standing platform. This subsystem also aims to provide real-time audio and visual feedback to the patient and therapist, and recording rehabilitation-session information for subsequent review.

Conventional balance assessment may rely heavily on therapist observation or standalone clinical equipment. Although direct clinical supervision remains important, the proposed subsystem introduces a low-cost embedded monitoring approach that is capable of continuously measuring bilateral loading while presenting the information both locally and through a web-based monitoring interface. The system therefore provides an additional quantitative indication of whether the user is maintaining a symmetrical loading or shifting excessive weight toward one side.

Previous studies have demonstrated the potential value of sensor-based biofeedback for balance rehabilitation. Force and inertial sensors have been used to monitor static and dynamic balance, while visual and auditory feedback can supplement the user's sensory information during balance training. Research also reported that with audio-visual feedback, it can assist in reducing asymmetrical lower-limb loading in rehabilitation applications (*De Nunzio et al., 2014*).

The developed subsystem consists of primarily four 50 kg half-bridge load cells, two HX711 load-cell analogue-to-digital converter (ADC) modules, an ESP32 microcontroller, a 0.96-inch OLED display, a DFPlayer Mini audio module, a 4Ω 3W speaker, and a browser-based graphical user interface. The ESP32 performs embedded sensing and control functions, while the browser-based interface provides high-level data visualization, session control, data recording, and rehabilitation progress assessment.

The design fulfills the assignment requirement for a microcontroller-based control system that is capable of monitoring input parameters, processing the acquired data, controlling output devices, and presenting the system condition through the display.

1.1 Input/Output Interface Design and Program Development

1.1.1 Subsystem Overview

The main function of the Balance Rehabilitation Monitoring and Control Subsystem is to obtain quantitative information about the patient's left and right side weight distribution, while he/she stands on the rehabilitation platform. The measured forces are processed to determine the proportion of the total load carried by each side. Then, the resulting information will be presented to the patient and the therapist via visual and audio outputs.

The balance rehabilitation process can be represented by using the following signal flow:

The four half-bridge load cells are positioned under the platform so that two load cells contribute to the left-side measurement and the remaining two contribute to the right-side measurement. Each pair is connected to an individual HX711 module, which is responsible for left and right side processing. By connecting the load cells in this way, it enables the system to acquire left and right side loading independently.

The ESP32 was selected as the main controller because the system requires both conventional embedded interfaces and wireless connectivity. The ESP32 integrates Wi-Fi capability and supports peripherals such as *UART* and *I²C*, making it suitable for communicating with the OLED display, DFPlayer Mini, HX711 modules, and browser-based monitoring system. Espressif specifies that the ESP32 family provides integrated Wi-Fi and several digital communication peripherals, including *UART* and *I²C* (*Espressif Documentation*, 2025).

The subsystem contains three functional layers. The first layer is the sensing layer, which consists of load cells and HX711 modules. The second layer being the embedded control layer, consists of the ESP32, OLED display, and DFPlayer Mini. The third layer is the monitoring and application layer, which consists of the browser-based user interface that is mainly used for session management, real-time data presentation, historical records, graphical analysis, and rehabilitation trend assessment.

This distributed architecture was adopted after development testing showed that concentrating sensor acquisition, web communication, database operation, display updates, and higher level calculations on the ESP32 increase the likelihood of communication delays and system instability. Consequently, the final design retained time-critical sensor and control functions on the ESP32 while assigning more computationally intensive visualization and historical-data functions to the browser.

The major input and output elements are therefore summarized in the following table.

Table 1.1: Summary of all input and output in the system.

<i>Category</i>	<i>Component</i>	<i>Main Function</i>
<i>Input</i>	4 × 50 kg half-bridge load cells	Measure left and right applied load
<i>Input interface</i>	2 × HX711 ADC modules	Amplify and digitize load-cell signals
<i>Processing</i>	ESP32	Embedded sensing, control, and communication
<i>Local display</i>	0.96-inch OLED	Display operational information
<i>Audio output</i>	DFPlayer Mini + speaker	Provide verbal corrective feedback
<i>Interface</i>	Web application	Monitoring, control, and data analysis
<i>Database</i>	Firebase Realtime Database	Persistent patient / session storage

1.1.2 Hardware and Electronic Interface Design

The balance rehabilitation monitoring and control subsystem required both hardware and software to make this subsystem a perfect duo. Hardware is used to capture live data from the patient, whereas the software is used to perform wireless control, session monitoring, and also data logging and historical assessment. Each of these will be explained in detailed in the following sections.

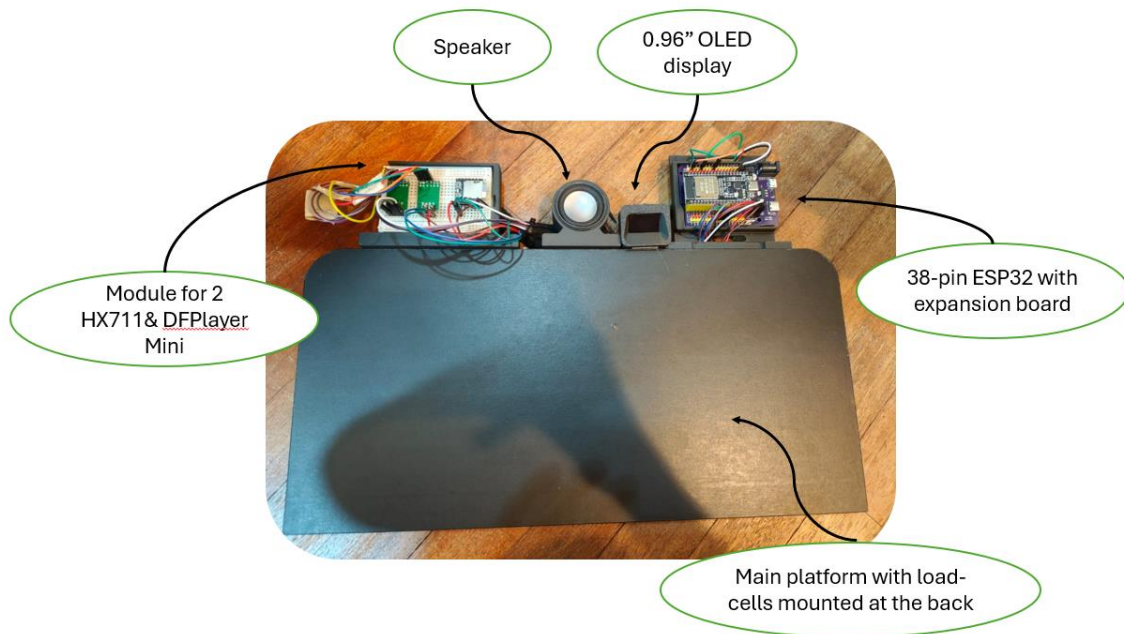


Figure 1.1: Overall hardware setup of the Balance Rehabilitation Platform.

1.1.2.1 ESP32 Microcontroller

The ESP32 serves as the central embedded controller of the balance rehabilitation subsystem. It acts as the “brain” for the system whereby it reads input from the sensors, and sends signals to the actuator. It acquires readings from the two HX711 ADC modules, from which it will perform necessary signal processing, manages the session state, updates the OLED display, activates the appropriate audio prompt, and also provides measured data to the browser-based user interface.

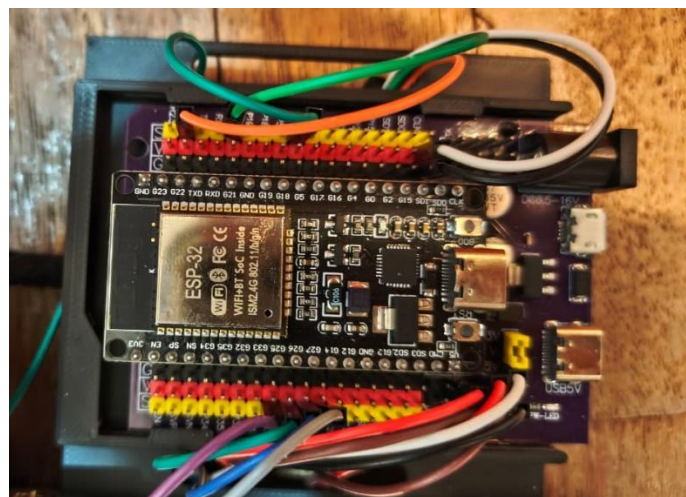


Figure 1.2: A 38-pin ESP32 connected in an expansion board.

The integrated Wi-Fi capabilities of the ESP32 was especially important because the project required online data access, communicate with browser-based interface, and control without

an additional wireless communication module. In return, this can reduce component count, wiring complexity, cost, and also system size.

The ESP32 also provides sufficient communication interfaces to support the required peripherals. The OLED display communicates using I^2C , whereas the DFPlayer Mini is controlled via *UART* communication. The load-cell interface uses digital data and clock signals from the HX711 modules.

The ESP32 performs the following important functions in this subsystem:

- a. Acquisition of left and right load-cell data;
- b. Communication with both HX711 modules;
- c. Communication with the OLED display;
- d. Communication with the DFPlayer Mini;
- e. Session-state management;
- f. Local feedback control;
- g. Wi-Fi connectivity;
- h. HTTP web-server operations; and
- i. Provide real-time JSON data to the webpage.

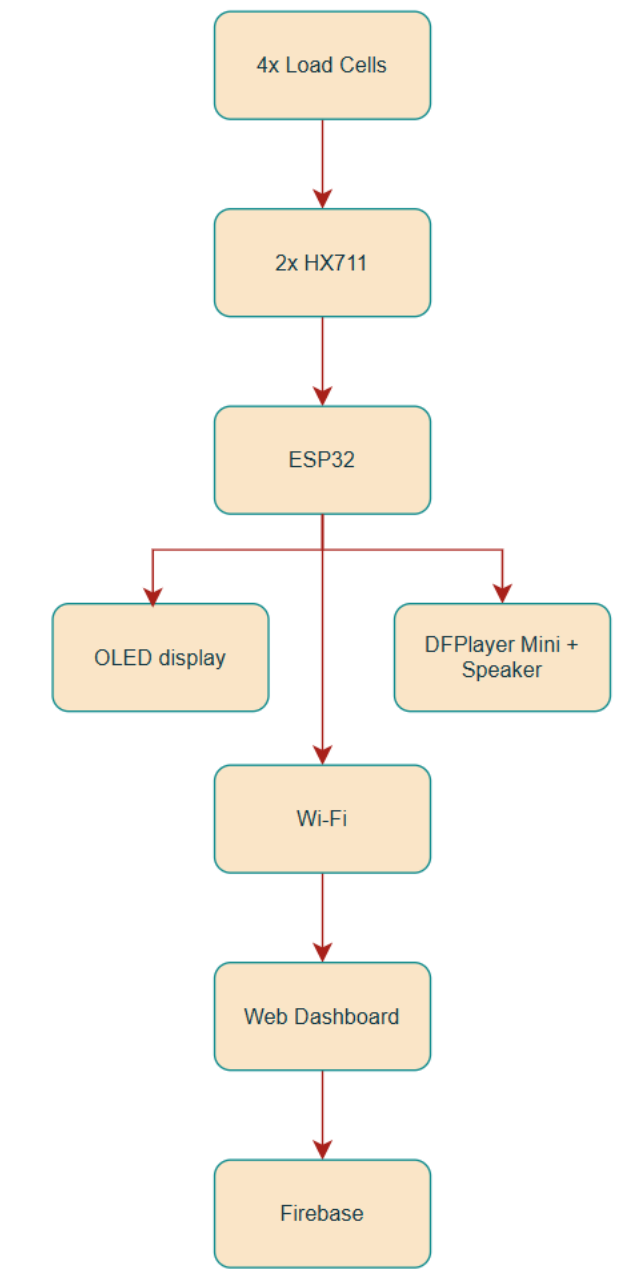


Figure 1.3: Hardware architecture of the Balance Rehabilitation subsystem.

A summary of the final hardware interfaces is as shown in the table below.

Table 1.2: Main electronic interfaces of the balance rehabilitation subsystem.

<i>Component</i>	<i>Interface</i>	<i>ESP32 Connection</i>	<i>Function</i>
<i>Left HX711 DT</i>	Digital	GPIO 25	Left-side load data
<i>Left HX711 SCK</i>	Digital	GPIO 14	Left HX711 clock

<i>Right HX711 DT</i>	Digital	GPIO 27	Right-side load data
<i>Right HX711 SCK</i>	Digital	GPIO 26	Right HX711 clock
<i>0.96" OLED SDA</i>	I^2C	GPIO 21	Display data
<i>0.96" OLED SCL</i>	I^2C	GPIO 22	Display clock
<i>DFPlayer Communication</i>	UART	GPIO 17	Audio module communication

The use of GPIO 12 initially introduced an additional design consideration because some ESP32 pins can influence the device boot state. During deployment, boot-related problems were observed when unsuitable peripherals were connected to boot-sensitive pins of the ESP32. This required careful allocation of peripherals connections and formed a part of the troubleshooting process, and it will be discussed in later section.

1.1.2.2 Load Cell Interface

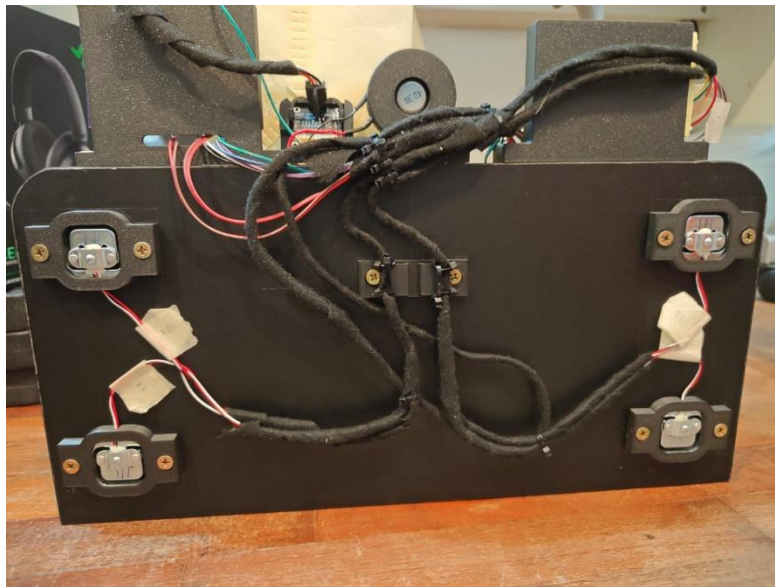


Figure 1.4: Arrangement of the four 50kg half-bridge load cells below the platform.

A total of four 50 kg half-bridge load cells were used to detect the force applied to the rehabilitation platform. Half-bridge load cells produce very small differential electrical signals proportional to the applied mechanical strain. These signals are too small to be measured directly by a conventional microcontroller digital input.

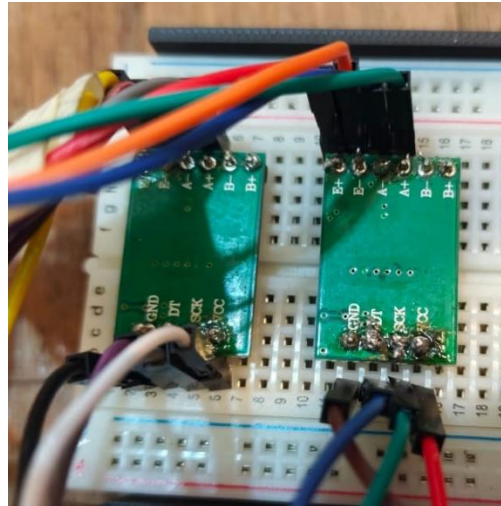


Figure 1.5: two HX711 ADC modules used in the platform.

To cope with this issue, two HX711 modules were introduced into the circuit. The HX711 is designed for precision weighing applications and provides amplification together with high-resolution analogue-to-digital conversion. One HX711 module was dedicated to the left pair of the load cells, while the second one was dedicated to the right pair of the load cells. This setup will ease the work of balance calculations when the patient is standing on the platform.

Using this particular setup, it shall provide two independent measurements, as stated below:

$$W_L = \text{Left side measured load}$$

$$W_R = \text{Right side measured load}$$

Hence, the total measured load can be obtained from:

$$W_T = W_L + W_R$$

Eq. 1.1

where:

W_T = total measured load,

W_L = left-side load, and

W_R = right-side load.

It is actually necessary to use separated measurement channels because the objective of this balance rehabilitation subsystem is not merely measuring the total body weight. Instead, the relevant rehabilitation information actually comes from measuring the weight distribution between the left and right sides.

1.1.2.3 Load Cell Calibration and Tare

Each of the HX711 channel requires calibration before meaningful weight measurement can be obtained from it. The raw digital output depends on the sensitivity of the individual load cells, mechanical installation, wiring, and amplifier characteristics. Consequently, a known reference load was used during the calibration process. Hence, the general conversion relationship can be expressed as the equation below:

$$W = \frac{R - R_0}{C_f}$$

Eq. 1.2

where:

W = calculated weight,

R = current raw reading,

R_0 = tare or zero-load offset, and

C_f = calibration factor.

The tare process establishes the zero reference when there is no external load applied to the platform. Without taring, the weight of the platform itself and small electrical offsets would be included in the subsequent readings, which will affect the accuracy of calibration and its subsequent measurement.

Also, both the left and right channels were calibrated separately because their mechanical arrangement and sensor responses were not assumed to be identical. This approach also enabled correction of differences between the two sides.

Calibration was tested using known loading conditions during development. The calibration process is done by using a known weight (from a dumbbell) to obtain the calibration factor. The final calibration factors used in the completed program were:

- *Left* $C_f = -45580.0$
- *Right* $C_f = -41675.0$

The calibration routine improved the agreement between the measured values and the known applied loads.

1.1.2.4 0.96-Inch OLED Display

A 0.96-inch OLED display module was also integrated into the subsystem to provide immediate local feedback, aside from the auditory feedback. The display communicates with the ESP32 via the I^2C interface using SDA and SCL connections.



Figure 1.6: The 0.96-inch OLED display secured in a 3D-printed holder.

The OLED allows important status information to remain available even when the browser-based user interface is not being viewed. Depending on the current operating state, the display is used to present useful information such as system readiness, session condition, weight distribution status, and any relevant user prompt.

A local display was retained even though the project already included an online interface because an embedded rehabilitation system should not solely depend on a remote web page for basic operational feedback. Another important thing is that, in order to reduce unnecessary processing load, the OLED update rate was subsequently reduced instead of keep refreshing continuously at the maximum refresh rate.

1.1.2.5 DFPlayer Mini and Speaker Interface



Figure 1.7: A DFPlayer Mini module used to amplify audio tracks to the speaker.



Figure 1.8: A 4 Ω 3W speaker for playing real-time auditory feedback.

A DFPlayer Mini module and a 4 Ω 3W speaker were also incorporated into the balance rehabilitation platform to provide real-time auditory feedback to the patient. This allows the system to communicate prompts without requiring the patient to keep looking at the display, which is somehow improving the product's ergonomic value.

The module communicates with the ESP32 via *UART*. GPIO 17 was used for serial communication in the final stable configuration. Also, a 1 k Ω series resistor was added into the communication path to improve the compatibility with the DFPlayer input.

The stored audio tracks were assigned to individual operating conditions, which includes prompts for the following:

- a. Beginning a rehabilitation session;
- b. Maintaining or returning towards the center of the platform;
- c. Correcting excessive left-side load;

- d. Correcting excessive right-side load;
- e. Ending the session; and
- f. The implemented sleep or platform inactivity reminder.

Audio feedback was included because rehabilitation feedback may be easier to follow when the patient can give his/her focus and maintain his/her attention on body posture instead of repeatedly looking at a screen. Biofeedback research has also reported that auditory and visual feedback can both be used to augment balance-related sensory information (*Ma et al., 2016*).

1.1.2.6 Power Supply

During initial development, the ESP32 and the connected peripherals were powered only using a USB connection sourced from the computer. However, intermittent resets and brownout conditions were encountered when the complete system was operated.

After several testing were conducted, it indicated that the stability of the supply cable and power source has significant effects on the system. The USB cable was therefore replaced with a more reliable cable and a dedicated 5V phone charger was subsequently used to power the system. The ESP32 expansion board was also powered using a stable external supply.

This modification contributes in reducing the amounts of unexpected resets while demonstrating that the power subsystem must be considered as part of the complete embedded design rather than assuming that any USB source is able to provide sufficient and stable power for operation.

1.1.3 Program Development and Balance Monitoring Logic

The software for the subsystem was developed primarily in Arduino-compatible C++ for the ESP32 itself, and paired with JavaScript for the browser interface. The embedded program is responsible for acquiring physical measurements and controlling local devices, whereas the browser code is responsible for performing user-interface and higher-level session functions.

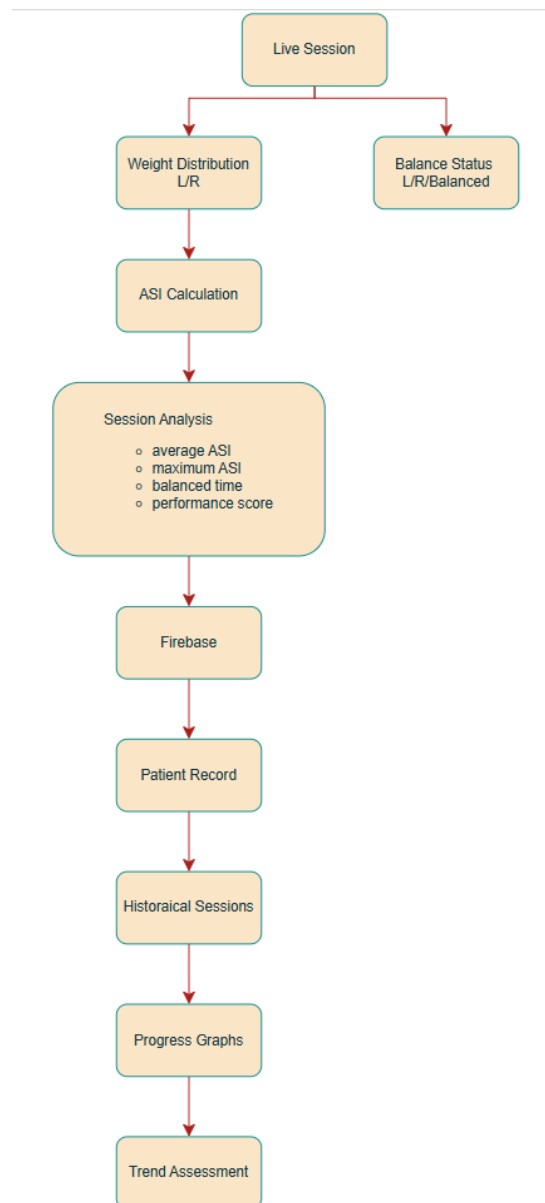


Figure 1.9: The software architecture of the balance rehabilitation subsystem.

In this particular assignment, it is required that the individual subsystem program to read input data, process the information obtained, and control output devices. The developed software indeed follow this input-process-output structure.

1.1.3.1 System Initialization

When power is supplied to the ESP32, the program performs the following initialization sequence:

1. Serial communication is started for debugging;
2. The left and right HX711 modules are all initialized;
3. The OLED display is configured;

4. The DFPlayer Mini communication interface is initialized;
5. Wi-Fi connection is established;
6. HTTP server initialization; and
7. The system enters the idle state.

The program will check whether the HX711 modules are ready before attempting to read the real-time data. This precaution step can avoid forcing the controller to remain blocked indefinitely while waiting for a sensor response.

1.1.3.2 Load Cell Acquisition and Processing

During system operation, the ESP32 will continuously check on both HX711 modules. When valid readings are available, the raw values of the load cells are converted into the calibrated values of left and right loads. As a result, the values are stored as W_L and W_R , which represents the measured left and right loads respectively.

Also, to prevent any unnecessary triggers, a minimum total-load condition is introduced into the program before the rehabilitation calculations are considered valid. This can prevent readings produced when no patient is standing on the platform from being interpreted as genuine rehabilitation data, which can significantly affect the result accuracy.

In the final system, a total load of 5 kg is used as the minimum valid load condition before the balance statistics are being accumulated. This is crucial as it could prevent noise and near-zero readings from incorrectly affecting the ASI and session statistics, which will be recorded for the patient history.

1.1.3.3 Weight Distribution Calculation

The measured left and right values shall allow the relative load percentages to be calculated.

The left-side percentage is:

$$P_L = \frac{W_L}{W_L + W_R} \times 100$$

Eq. 1.3

The right-side percentage is:

$$P_R = \frac{W_R}{W_L + W_R} \times 100$$

Eq. 1.4

where:

P_L = left-side load percentage, and

P_R = right-side load percentage.

If the patient is under an approximately symmetrical standing condition, then $P_L \approx 50\%$ and $P_R \approx 50\%$. If there is a shift towards the left, it shall increase P_L while reducing P_R at the same time. And the same goes to the opposite case, which will produce a mirrored response of this. The system therefore interprets the difference between left and right load distribution as an indication of balance asymmetry.

One simpler expression for this absolute difference is as follows:

$$E_B = |P_L - P_R|$$

Eq. 1.5

where E_B represents the bilateral load imbalance.

The final stability score presented on the website is also calculated according to the implemented browser algorithm.

1.1.3.4 Absolute Symmetry Index (ASI)

The Absolute Symmetry Index (ASI) is the principal quantitative indicator that is used in the developed balance rehabilitation system. It is a measurement that determines whether the patient's balance is acceptable via the received data input from the load cells.

Instead of considering only the difference between the two weights, ASI actually express the bilateral asymmetry relative to the combined loading. The calculation of ASI is expressed as:

$$ASI = 2 \left| \frac{W_L - W_R}{W_L + W_R} \right| \times 100\%$$

Eq. 1.6

where W_L and W_R represent the measured weight on the left and right sides of the platform.

By referring to Equation 1.6, it can be comprehended that when an ASI value of:

$$ASI = 0\%$$

represents a perfect left-right symmetry. Hence, as the difference between both sides increases, the ASI also increases accordingly.

For example, if the patient's stance is imbalanced, and his/her weight tends to shift towards either side of the platform, then the value of the ASI will increase. In contrary, if the patient's stance approaches an equal 50:50 distribution, the ASI will approach zero. Therefore, this index is very suitable for determining the patient's balancing status as it provides a single numerical indicator that is easier to compare across repeated samples and rehabilitation sessions.

For the developed final prototype, an ASI of approximately:

$$ASI \leq 10\%$$

is treated as the defined balanced region. Note that the 10% boundary is used as an engineering threshold for the prototype's feedback and performance evaluation. It should be taken as a universal clinical diagnostic threshold. Also, only samples where the total measured weight exceeding the minimum threshold are included in the session calculations. Other small drifting will be neglected to avoid miscalculation and malpresentation of data. This step avoids invalid ASI calculations when the platform itself is unoccupied.

The current ASI is continuously recalculated during an active rehabilitation session, and this data forms the backbone for:

- a. Patient's balance status;
- b. Average ASI;
- c. Maximum ASI;
- d. Balanced-time percentage;
- e. Session performance;
- f. Audio feedback; and
- g. Longitudinal rehabilitation assessment.

ASI therefore acts as the central link between the physical sensing subsystem and the higher-level rehabilitation data analytics.

1.1.3.5 Balanced-Time and Performance Calculation

There are several rehabilitation metrics that are being calculated throughout the whole active session. The first metric is the average ASI:

$$ASI_{avg} = \frac{\sum_{i=1}^n ASI_i}{n}$$

Eq. 1.7

where n represents the number of valid samples during the particular session. The average ASI is able to provide a better representation of the overall session than a single instantaneous reading, due to the fact that the patient's balance may fluctuate throughout the rehabilitation exercise.

Meanwhile, the system will also record the maximum ASI:

$$ASI_{max} = \max(ASI_1, ASI_2, \dots, ASI_n)$$

Eq. 1.8

This metric helps to identify the largest asymmetry observed during the whole active session.

The third metric that is measured in the session is the balanced time. Each of the valid samples is examined to determine whether they fulfilled the threshold of:

$$ASI \leq 10\%$$

If so, then the particular sample is classified as balanced. Hence, the percentage of the session spent in the balanced region is therefore expressed as:

$$Balanced\ Time(\%) = \frac{N_{balanced}}{N_{valid}} \times 100\%$$

Eq. 1.9

where:

$N_{balanced}$ = number of valid samples that is within the balanced region; and

N_{valid} = total number of valid samples in the session.

Hence, it can be comprehended that a patient who is able to maintain symmetrical loading consistently, will therefore obtain a higher balanced-time percentage.

Furthermore, the webpage will also convert the average ASI into a intuitive session-performance score:

$$Performance = clamp(100 - ASI_{avg}, 0, 100)$$

Eq. 1.10

A lower average ASI will produce a higher performance score, since lower ASI means better balance. For better understanding of this particular equation, please refer to the following example.

Assuming that the patient obtained an $ASI_{avg} = 8\%$, and when incorporated into Eq. 1.10, the patient would therefore obtain a performance score of $Performance = 92$. The objective of this performance score is just for the purpose of easier visualization, and it is just a simple engineering indicator for the prototype itself. It should not be taken as clinically validated medical score.

1.1.3.6 Audio Feedback and Control Logic

The ASI and left-right loading condition are also used to determine the corresponding auditory feedback that should be given to the patient. If the patient remains within the defined range of symmetrical distribution, then the system will recognize the posture as balanced.

In contrast, if there is excessive load measured towards either side of the platform, the system will identify the direction of the asymmetry that occurred using the underlying left and right measurements. With this clear logic in mind, the corresponding audio prompt can then be triggered accordingly.

The general feedback logic is tabulated as follows:

Table 1.3: Audio feedback logic with respect to the patient's weight distribution.

<i>Measurement</i>	<i>Status</i>	<i>Audio Logic</i>
$ASI \leq 10\%$	Balanced	Maintain current position
$W_L > W_R$	Lean Left	Provide corrective instruction to the patient, more towards right
$W_R > W_L$	Lean Right	Provide corrective instruction to the patient, more towards left

The implementation also includes start, stop, calibrate, center, corrective, and sleep/inactivity audio prompts. The feedback is also controlled so that the same message will not be triggered continuously.

1.1.3.7 ESP32 HTTP Communication

The ESP32 operates as a light-weight HTTP server for the browser-based application. There are several endpoints used to exchange information and control the rehabilitation sessions. One of the most important endpoints is:

/data

This endpoint returns the current measurement and operational state in JSON format. The returned information includes:

There are also several other important endpoints that are used in the subsystem. All of them are tabulated as follows:

Table 1.4: HTTP endpoints and their respective functions.

<i>Endpoint</i>	<i>Function</i>
<i>/data</i>	Returns current measurements and operational state in JSON.
<i>/start</i>	Changes the embedded system into the active rehabilitation session state.
<i>/stop</i>	Terminate the current rehabilitation session.
<i>/tare</i>	Performs the zeroing procedure for the weighing system.

This HTTP architecture enables standard browser technologies to interact with the ESP32 without the need of a dedicated proprietary computer software.

1.1.4 Web-Based User Interface and Data Management

In this subsystem, the web application is one of the main functional components of the Balance Rehabilitation Monitoring and Control Subsystem. It was developed not simply as a display, it also acts as the primary monitoring, control, session-management, database-access, and rehabilitation-analysis interface.

The web application was developed and implemented using the combination of HTML, CSS, and JavaScript. The software communicates with the ESP32 for real-time physical information, while the Firebase Realtime Database provides a persistent online storage. The final architecture divides the system responsibility into three different levels as shown in the table below.

Table 1.5: Three levels of the system architecture.

<i>Layer</i>	<i>Main Responsibility</i>
<i>ESP32</i>	Real-time sensing and hardware control
<i>Web application</i>	Monitoring, calculation, control, visualization, and analysis
<i>Firebase</i>	Persistent patient and session data storage

This architecture is way more efficient, and it could significantly reduce the amount of processing and network workload required directly on the ESP32.

1.1.4.1 Live ESP32 Connection and Data Acquisition

The browser will access the /data endpoint of the ESP32 periodically. And the ESP32 will return the current sensor and operating information as JSON data. Next, the JavaScript in the browser will receive the JSON response from the endpoint, and then it will update the displayed values.

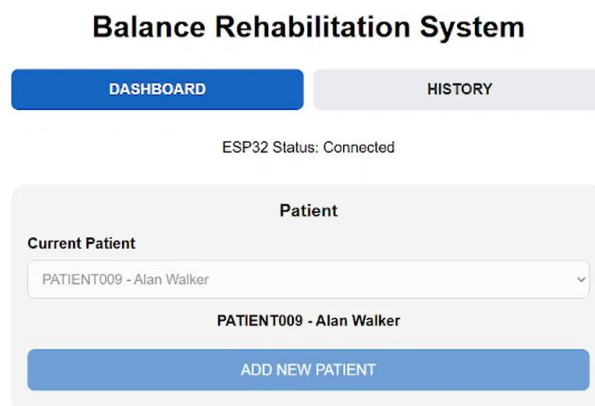


Figure 1.10: The web interface will display the ESP32 connection status.

The interface also provides a connection-status indication. This is very crucial because the therapist must be able to differentiate between:

A visible connection indicator therefore significantly improves the usability and operational safety.

1.1.4.2 Real-Time Weight Distribution

The web interface also displays the patient's current left and right load values. The corresponding percentages are also being calculated and displayed.

With this interface, it will be easier for the therapist to interpret and understand at a glance compared to the raw load-cell ADC output. The live distribution will also change based on the patient's body weight shifting across the balance platform.

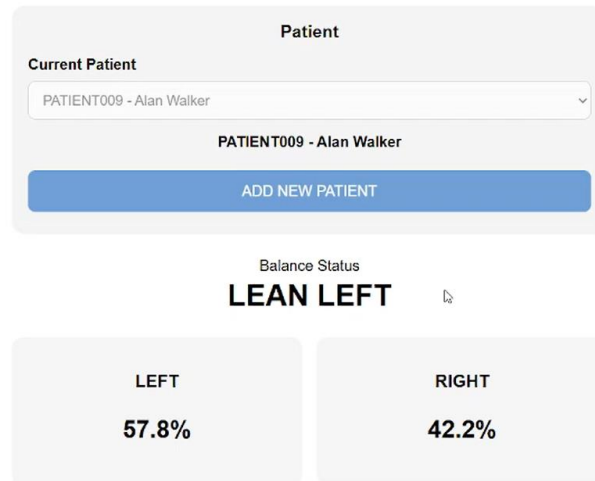


Figure 1.11: Real-time weight distribution of the patient is displayed.

In simple words, the web interface plays a role as real-time visual biofeedback system. With this interface, the therapist can immediately observe the patient's status, and can see whether he/she is approximately centered or leaning towards one side. With live readings available, it provides a higher accuracy and standardized factual output rather than solely relying on the therapist's eye leveling, which might vary from person to person (*Alhasan et al., 2017*).

1.1.4.3 ASI and Balance Monitoring

In addition to left and right percentages, the webpage also displays the current ASI of the patient. This is one of the core differences between the developed interface and a simple electronic weighing platform. In the developed system, the browser will track the following:

- Current ASI;
- Average ASI;
- Maximum ASI;
- Balanced-time percentage;
- Number of valid samples; and
- Session performance.

Current ASI provides information about the patient's instantaneous symmetry, whereas the average ASI provides information about the overall quality of the session. Also, the maximum

ASI identifies the worst observed imbalances. Meanwhile, the balanced time shows how consistently the patient remained within the desired region.

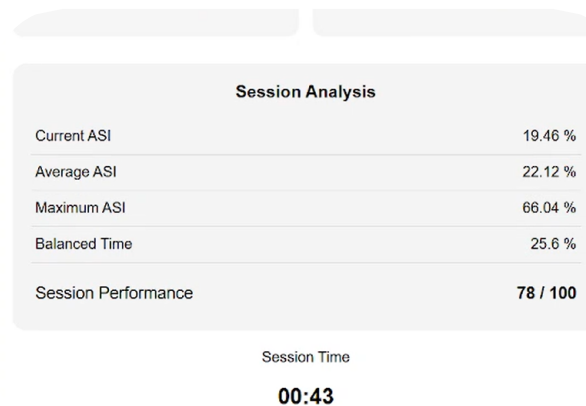


Figure 1.12: Evaluation metrics calculated during the active session.

Together, these indicators provide a more comprehensive and precise representation of the patient's balance performance compared to just simply displaying the patient's final position. These metrics are also useful in helping the patient to track and analyze the progression of his/her rehabilitation, rather than just displaying data without utilization (*Kim et al., 2022*).

For instance, two rehabilitation sessions may both end at a 50:50 distribution. However, in the first session, the patient may have remained closed to symmetrical distribution for almost the entire session. But in the second session, the patient may have spent most of the exercise heavily leaning towards one side before correcting his/her posture only at the end of the session. In this instance, if only the final measurement were considered, then both sessions could appear identical to the therapist.

To cope with issue, average ASI and balanced-time percentage allow these sessions to be differentiated. This is one of the primary reasons ASI-based session statistics were included in the web interface.

1.1.4.4 Session Control and Timer

The webpage also provides control functions that allow the rehabilitation operator to interact directly with the embedded system. The main functions of the web control includes:

- Strat Session
- End Session
- Calibrate/Tare

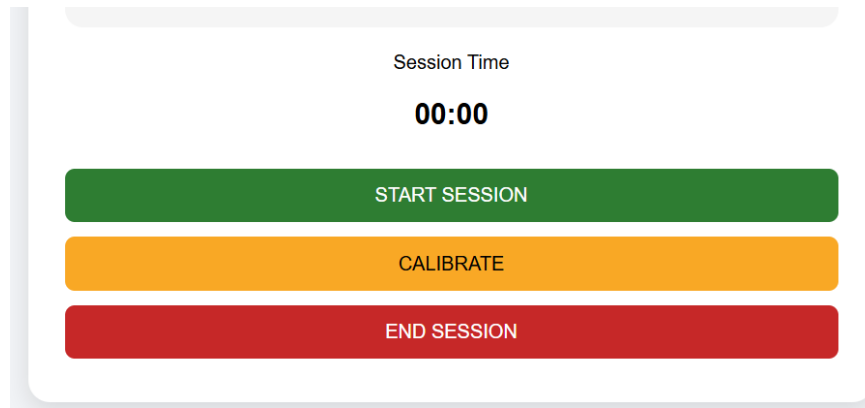


Figure 1.13: Session control and timer in the web interface.

When Start Session is selected, the ESP32 changes to the active-session state and the webpage begins recording and capturing session statistics. Tailing this, the session timer also begins.

During an active session, valid ASI samples are accumulated and are used to calculate the rehabilitation metrics. The calculated metric will be displayed on the web interface, which allows the therapist to interpret the data in a straightforward manner.

When End Session is selected:

- Data collection stops;
- Final session statistics are calculated;
- The session summary is generated; and
- The relevant data can be stored in Firebase.

During development, an issue was identified where the session button and timer did not reset correctly when a second rehabilitation session was started. Following this problem, the JavaScript session-state logic was subsequently updated and corrected so that the consecutive sessions could be initialized properly.

1.1.4.5 Session Performance Summary

After the rehabilitation exercise ends, the webpage will produce a summary of the completed session. The generated session summary will contain information such as:

- Session duration;
- Average ASI;
- Maximum ASI;
- Balanced-time percentage;
- Performance score; and

- Number of valid samples.

With this feature enabled, it can provide a concise overview of the patient's performance. The summary also allows the therapist to evaluate the complete exercise rather than solely relying on instantaneous observations during the session.

For example, a reduction in average ASI between sessions means an improvement in bilateral symmetry, while an increase in balanced-time percentage means an improvement in the patient's ability to maintain a symmetrical weight distribution.

1.1.4.6 Firebase Realtime Database

Firebase Realtime Database was incorporated into the system to provide persistent online data storage. Without a database, the rehabilitation measurements would be lost once the browser was closed or the current session is ended. With Firebase used as the data storage, it enables completed session records to be stored and retrieved later when needed.

The database can store information such as:

- a. Patient identification;
- b. Date and time of the session;
- c. Session duration;
- d. Average ASI;
- e. Maximum ASI;
- f. Balanced-time percentage;
- g. Performance score;
- h. Valid sample counts; and
- i. Other relevant rehabilitation data.

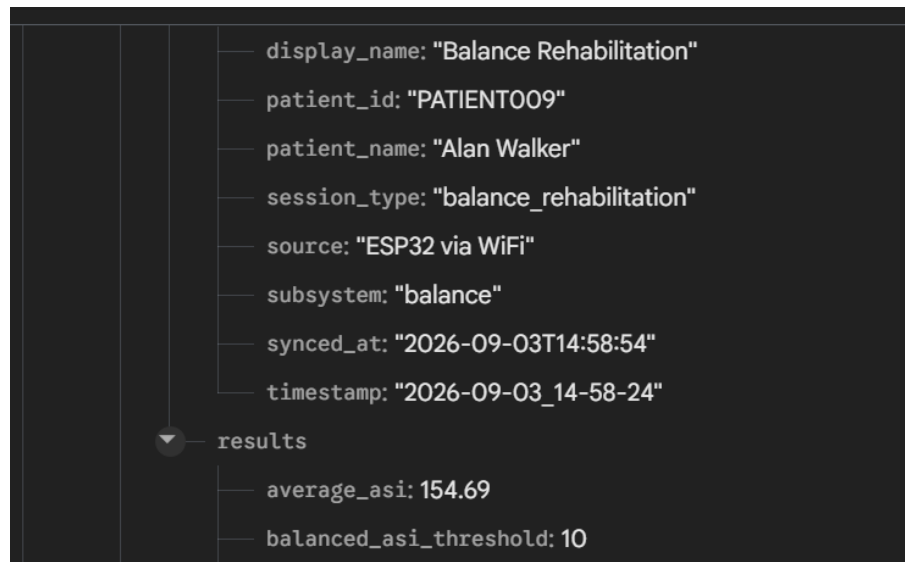


Figure 1.14: One of the session data stored in the Firebase for PATIENT009.

The database structure separates the records according to patient. This arrangement allows each patient's rehabilitation history to be retrieved independently, without mixing up with each other.

1.1.4.7 Patient and Session Management

Patient-specific data management was implemented because rehabilitation progress should be evaluated for an individual rather than combining all users.

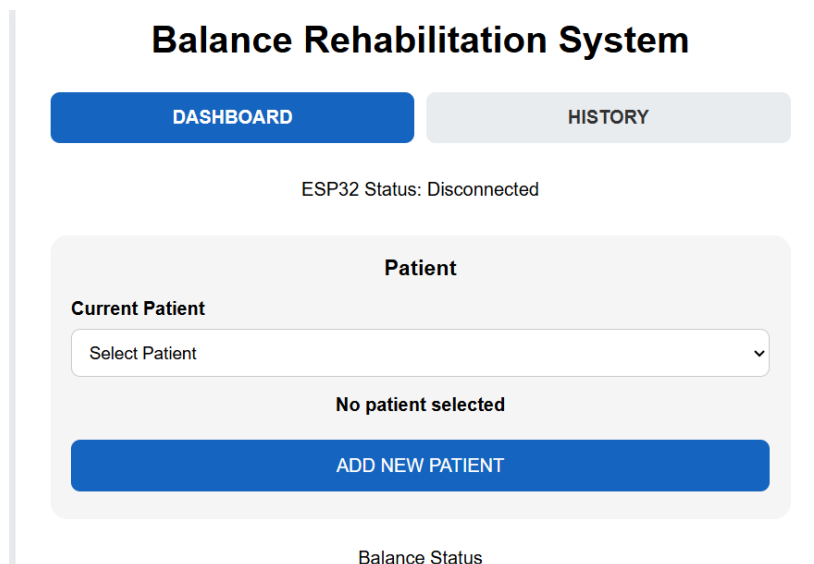


Figure 1.15: Patient has to be selected before starting a session or acquiring session history.

When a patient is selected, the webpage can associate newly completed sessions with that particular patient's database record. The system is able to retrieve the corresponding historical sessions. This allows the same hardware platform to be used by multiple patients while also maintaining separated rehabilitation records.

This is crucial as the patients' progression and their respective data should be remained private and confidential, which means that mixing up with different patient records would cause significant negative impact. Each patient's record therefore represents an independent series of rehabilitation sessions. This feature improves the practical value of the system because it transforms the platform from a simple one-session prototype into a basic rehabilitation monitoring system.

1.1.4.8 Historical Records and Progress Graphs

Historical session information is retrieved from Firebase and is presented via the webpage. Instead of requiring the therapist to inspect the individual database records manually, the information is displayed in a more readable way.

For each completed session, relevant parameters can be presented, which includes the following:

- a. Average ASI;
- b. Maximum ASI;
- c. Balanced time;
- d. Performance score; and
- e. Session duration.

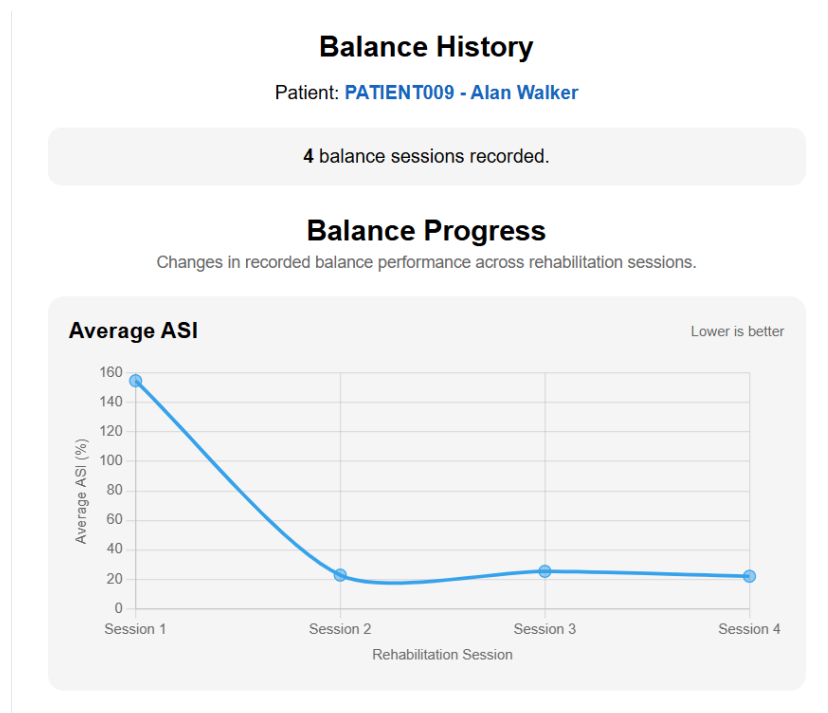


Figure 1.16: Average ASI graph generated from accessing PATIENT009.

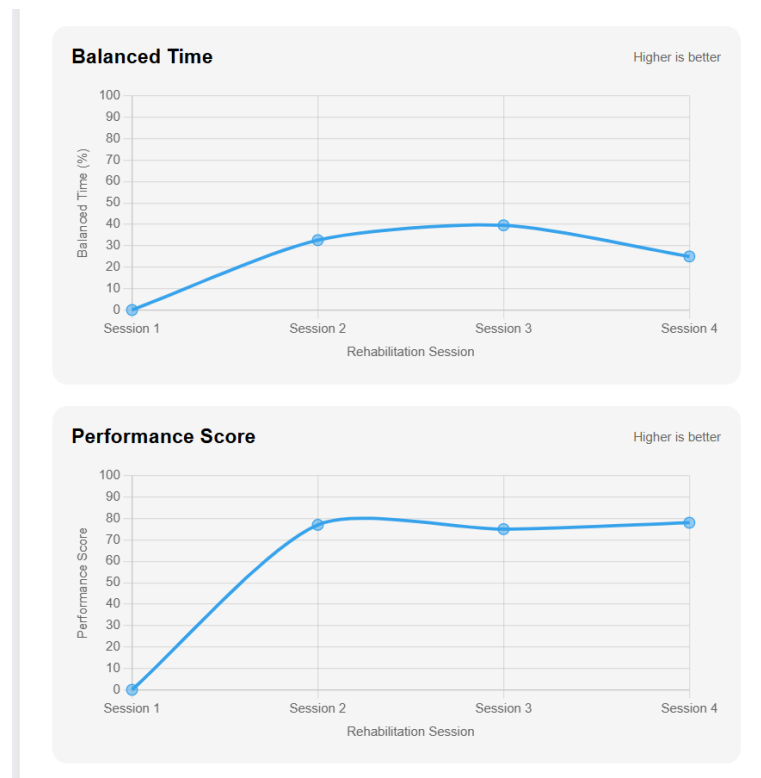


Figure 1.17: Balanced time and performance score graphs generated from PATIENT009's progression.

Since the data are stored, they are also used to generate graphical representation for the therapist and patient to view. The visual aids generated can help both the therapist and patient to understand better regarding the progression of the rehabilitation, instead of only looking at numbers.

For instance:

- Decreasing average ASI indicated an improvement in bilateral symmetry;
- Increasing balanced time means a more consistent balance;
- Increasing performance score indicates overall improvement according to the project-specific metric.

Hence, with this function added into the system, the data are converted into useful visual progression, which would definitely help the therapist and patient to understand the progression at one glance.

1.1.4.9 Rehabilitation Trend Assessment

Once there are sufficient patient-session information are collected, the webpage will perform a basic trend assessment. This function helps to analyze the performance of the patient based on his/her past data, and will generate a short summary of that particular patient accordingly.

The system was tested using at least three recorded sessions for one patient. The recent sessions results are compared to determine whether the patient's measured balance performance appears to be either:

- Improving;
- Relatively stable; or
- Deteriorating.

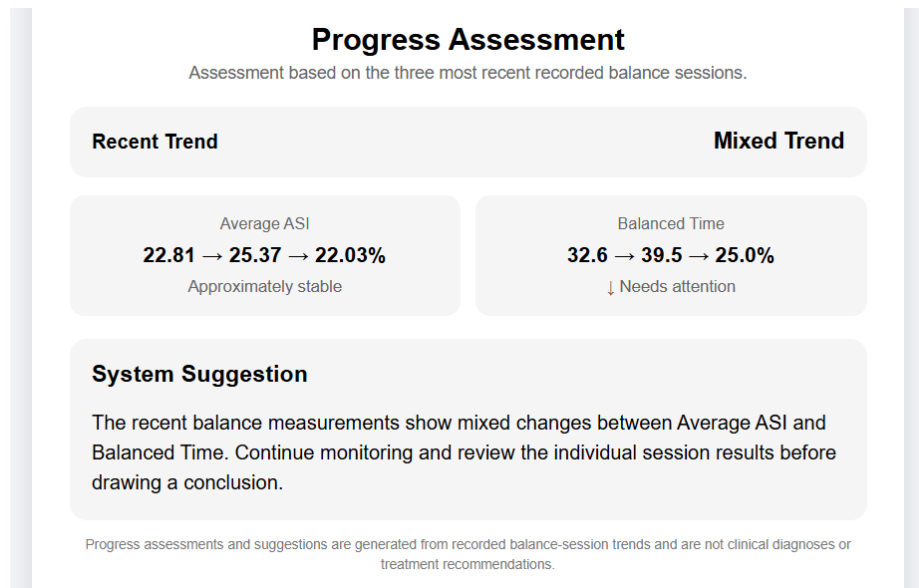


Figure 1.18: Progress assessment generated by the system based on past performance of PATIENT009.

The assessment helps to reduce the amount of manual comparison had to be done by the therapist when there are several session data available.

For instance, if the average ASI gradually decreases across multiple sessions conducted while balanced time increase, the system will recognize this as an improving trend, and will show the analyzed trend to the therapist.

However, this result should not be taken as a clinical diagnosis. The trend assessment is an engineering decision-support feature that is intended to help the therapist to review the recorded rehabilitation measurements. A qualified healthcare professional remains responsible for clinical interpretation.

1.1.5 Design Considerations

The design of this subsystem had come across different considerations, including public health and safety, whole-life cost, energy efficiency, resource efficiency, accessibility, societal impact, and environmental sustainability.

1.1.5.1 Public Health and Safety

Safety is one of the most important factors in the design consideration because the patient performs the rehabilitation exercise while standing on the platform. Therefore, the platform is designed in a way that it can hold itself up steadily on the ground. During operation, it should be placed on a stable and level surface before use.

The mechanical structure of the platform is also designed to be sufficiently rigid so that any unexpected platform movement will not cause the patient to fall, eliminates falling risk. Also, all the wires are cleared from the standing area, preventing the patient from falling.

Another thing is that all the electronics components are operating at a relatively low voltage (3.3V), which in turns reduce the electrical risk, although all the connection point are already insulated properly.

Moreover, calibration and tare are also safety-related functions. This is because calibrating and taring can avoid wrong sensor readings. Having correct sensor readings are important because any incorrect sensor measurements may provide misleading balance feedback, and it will directly cause wrong judgement from the therapist.

Most importantly, the developed system is a balance rehabilitation prototype rather than a certified medical device. Hence, it should be used under suitable supervision during the presentation.

1.1.5.2 Whole-Life Cost

The whole-life cost is also considered during the selection of the system architecture. The design of the system uses a widely abundant modular components, which includes:

- a. ESP32;
- b. HX711;
- c. Load cells;
- d. OLED;
- e. DFPlayer Mini; and

f. Standard speaker.

Since the ESP32 already contains Wi-Fi capabilities, this helps in eliminating the need for a separated wireless module. Similarly, the system also utilized a browser-based application for control and monitoring of the platform and patient status. And with this IoT available, it eliminates the need for dedicated commercial human-machine interface.

Also, since the components are abundant, and each part of the systems are separated, then they can be replaced independently if there is a failure occurring. This structure improves maintainability and can significantly reduce the replacement cost throughout the system's lifespan.

1.1.5.3 Energy Efficiency and Net-Zero Consideration

The system cannot reasonably be described as completely net-zero. Instead, the design of this subsystem aims to reduce unnecessary energy consumption. First of all, the OLED display used in the system is relatively small, only diagonally 0.96 inch. Smaller OLED therefore consumes less power than a large dedicated display. Next, the display refresh rate is also reduced to a relatively suitable level whereby it would not cause any unnecessary redraw operations, thus reducing energy consumption.

Moreover, the OLED display also incorporated "sleep mode", whereby the screen will turn off after a period of inactivity. When the platform is left empty for 1 minute or more, the screen will be turned off. This can definitely reduce energy consumption instead of let the screen keep running even there's no patient on the platform.

Another measure that is used in the system development is that, only the lightweight real-time operations are given to the ESP32. The purpose of this measure is to reduce the workload of the ESP32, thus reducing the power consumption of the microcontroller. For all the heavy tasks such as metric calculation, graphical representation, or historical data logging, these tasks are carried out on the web server itself whereby they have better chipset and is more energy efficient than directly assign these tasks to the ESP32 itself. By moving the higher-level processing to the existing computer browser, this can definitely reduce the requirement for additional dedicated hardware, as well as performing complex task on higher efficiency computer CPU will reduce energy consumption significantly.

All the considerations are done before building the prototype, which makes the execution phase more efficient. With lower energy consumption across all the fields, this indirectly contributed

to lower greenhouse gases (CO_2) produced, which aligns with the goal of achieving net-zero carbon emission.

1.1.5.4 Efficient Resource Usage

The efficient use of ESP32 processing and memory resources had a significant influence on the final system architecture. An earlier version of design attempted to perform more network and Firebase operations directly on the ESP32 itself. And this heavy workload on the ESP32 produced issues such as memory pressure, website communication delays, system instability, reset and reboot behaviour, and slower server response. These are all unwanted issues caused by unoptimized ESP32 processing and memory usage that would negatively affecting the user experience (UX) and it must be avoided.

Therefore, it was then redesigned into a more efficient architecture. In the final prototype and system, the ESP32 focuses on:

- Sensor acquisition;
- Hardware control;
- Real-time OLED output and audio feedback; and
- Lightweight HTTP responses.

while the browser handles:

- ASI related session analytics;
- Graphical interface;
- Session management;
- Patient's session history and chart;
- Firebase interaction; and
- Trend assessment.

This distributed approach uses the limited embedded resources more efficiently, and significantly improving the overall user experience (UX).

1.1.5.5 Accessibility and Societal Considerations

This subsystem provides both visual and auditory feedback. The visual interface helps the therapists and patients who prefer numerical or graphical information, while the audio prompts reduce the need for the patient to look at the OLED screen or monitor.

The use of recorded audio files also makes future multilingual support relatively straightforward. Additionally, this balance rehabilitation subsystem also has potential societal values as a lower-complexity tool for basic balance monitoring. Despite it cannot fully replace the advanced clinical force plates, this subsystem, however, demonstrates how relatively affordable sensing and web technologies can be integrated to provide objective-based rehabilitation and provides accurate measurements.

1.1.5.6 Environmental Sustainability

Last but not least, the modular design of this subsystem also contribute to sustainability because individual components can be replaced without discarding the entire system, which makes it like a “Lego”, where the user just have to dismantle the failed parts and replace it easily. For example, if the load cell is failing, then it can be easily replaced by simply unscrew the support at the base, and put in the new load cell. This makes future maintenance work relatively easier, which makes it more sustainable.

Furthermore, software-based improvements can also extend functionality without requiring new hardware. This makes future updates can be done easily with a relatively low cost. For example, a translation feature can be added into the system without much hassle. It can be done by just updating the audio library stored in the SD card, as well as updating the interface of the webpage.

Nevertheless, the electronic components will eventually come to the end of their own product life cycle, contributing to generating electronic waste. Therefore, appropriate reuse, repair, and proper disposal practices should be therefore considered in the future development.

1.2 Enhancement / Innovation of the Subsystem

There are also several enhancement and innovation incorporated into the subsystem via improved sensing, control logic, system functionality, or efficiency.

1.2.1 ASI-Based Balance Assessment

One of the most significant enhancement and innovation of the balance rehabilitation subsystem is the introduction of ASI-based balance assessment. This feature significantly improves the system compared with just simply displaying the left and right weight distribution readings. With this feature enabled, it will converts the raw readings captured into useful data that can be used for further assessments.

The raw weight measurements alone can only indicate how much force is applied to each sides of the balance platform, but these values alone do not provide a single numerical measure of symmetry. Hence, with ASI used as a metric, it combines both measurements into one normalized indicator that can be easily comprehended by both the therapist and the patient.

This feature allows:

- Real-time comparison;
- Balanced-region classification;
- Average-session analysis;
- Balanced-time calculations; and
- Longitudinal trend comparison.

Therefore, ASI converts the raw sensing system into a more meaningful rehabilitation monitoring platform. And from there, more advanced features such as graphical data display can also be done to help visualize the progress of the balance rehabilitation.

1.2.2 Web-Based Real-Time Monitoring and Control

The second major enhancement that is done in the subsystem is the browser-based user interface. Instead of requiring the therapist to stand directly next to the patient, he/she can remotely monitor the patient's status from a connected computer. Besides that, the therapist can also control the session start/stop via the webpage. This saves a lot of effort of the therapist and he/she can work remotely to monitor the patient without being present physically in the rehabilitation center. This concept is matching the Internet of Things (IoT), whereby the balance platform is now smart enough to connect to Wi-Fi and enabling wireless control over the network.

1.2.3 Firebase-Based Patient and Session Storage

With such advanced webpage control and data logging, there must be another "container" to store all the relevant information and session data of the patient. Firebase is therefore introduced into the subsystem, where it acts as the database for storing all the important information. This approach extends the subsystem to go beyond real-time monitoring.

With Firebase Realtime Database integrated into the balance rehabilitation subsystem, completed sessions can now be stored online and associated with individual patients. In simple words, it can now store the session data such as ASI, balanced-time, performance score, etc. according to which patient is performing that particular session. This allows rehabilitation information to remain available even after the balance rehabilitation exercise has ended.

This architecture then built the foundation for:

- Patient-specific records;
- Historical review;
- Remote data access;
- Graphical analysis; and
- Long-term progress monitoring.

1.2.4 Multi-Session Rehabilitation Trend Assessment

In addition to all the features above, there is also one more important innovation, which is the historical session data are also used to evaluate the patient's rehabilitation progression across several completed sessions.

Instead of assessing only the current exercise, the system can compare the current session with the previous performance. This feature is available because of the Firebase that is storing all the past session data of the patient. Then, the trend assessment can be done by pulling out the old data and comparing them with the current one. And from there, more and more features are generated such as graphical data plots for better visualization, and also trend assessment and summary based on the patient's past progress. The system will also provide a suggested number of future sessions required based on the performance of the patient by referring to his past data.

This feature is extremely important as rehabilitation is a long process, and improvement shall occur over a period of time. This trend-assessment feature therefore represents an enhancement from instantaneous monitoring towards longitudinal rehabilitation monitoring.

1.2.5 Comparison with Existing Rehabilitation Systems and Product Justification

Commercial rehabilitation systems such as dedicated force plates and balance-training platforms can provide substantially more sophisticated measurements than the developed

prototype. Such systems may measure the full center-of-pressure (COP) trajectories, dynamic postural stability, fall-risky parameters, and other clinically validated variables.

Therefore, the objective of the developed system is not to claim superior over the existing commercial clinical equipment. Instead, the prototype is trying to address a narrower requirement: accessible bilateral weight-distribution monitoring combined with real-time feedback, web-based control, online session storage, and longitudinal progress tracking.

Noticed that the proposed system actually occupies an intermediate position between a simple weighing platform and a complete clinical force-plate system. A conventional weighing device can measure body weight but provides very limited rehabilitation information. While a professional force-plate system provides a significantly more detailed clinical analysis but usually requires more specialized hardware and software. Then, the proposed balance rehabilitation subsystem actually fill up the sweet spot between these two. It focuses on the functions required for basic supervised bilateral balance rehabilitation with innovative IoT features, while maintaining a relatively low owning cost of the platform.

Table 1.6: Comparison between proposed system with existing products in the market.

<i>Feature</i>	<i>Proposed System</i>	<i>Advanced Commercial Balance System</i>
<i>Left/right loading</i>	Yes	Yes
<i>ASI/symmetry assessment</i>	Yes	May use more advanced metrics
<i>Real-time monitoring</i>	Yes	Yes
<i>Audio feedback</i>	Yes	Depends on product
<i>Web-based interface</i>	Yes	Depends on product
<i>Session history</i>	Yes	Yes
<i>Multi-session progress</i>	Yes	Yes

<i>Full center-of-pressure analysis</i>	No	Commonly available
<i>Dynamic force-plate assessment</i>	No	Commonly available
<i>Clinical validation</i>	No	Often available
<i>Modular electronics</i>	Yes	Primarily proprietary
<i>User-modified software</i>	Yes	Usually limited

By referring to the summary table above, it clearly shows that the proposed balance rehabilitation subsystem actually has several practical strengths. First of all, the system is focused on a very straightforward rehabilitation objective: identifying and reducing bilateral loading symmetry.

Secondly, the web interface allows monitoring and control without requiring a dedicated proprietary terminal. Third, this balance rehabilitation subsystem uses a modular architecture, which significantly reduces the overall system complexity by simplifying repairs, modifications, and future expansion.

Furthermore, the subsystem also incorporated Firebase Realtime Database, which allows the patient's data to be retained and reviewed across several rehabilitation sessions. And this is one of the main strengths of the system, where it allows the therapist to keep track of every progress of each patient with no hassle at all. In addition to the real-time database, the balance rehabilitation subsystem also incorporated the combination of measurable metrics such as ASI, balanced time, graphical history, and trend assessment. All these would allow the system to provide a more useful information than just giving an instantaneous left-right weight display.

Therefore, this product positioned itself as a low-complexity connected rehabilitation monitoring platform, rather than a replacement for high-end clinical balance systems. However, for detailed posturography, fall-risk diagnosis, or clinically validated neurological assessment, professional equipment remains a more appropriate option.

But for supervised basic weight-shifting and bilateral symmetry exercises, the proposed system demonstrates how relatively simple embedded hardware can provide objective-based monitoring, multimodal feedback, and longitudinal session analysis.

1.3 Results and Discussion

This section of the paper presents the analysis of results, development problems, troubleshooting, limitations, and achievement of individual objectives.

1.3.1 Subsystem Testing and Results

1.3.1.1 Load Cell and HX711 Testing

Each HX711 and load-cell channel was initially tested separately. The serial monitor was used to confirm whether the left and right HX711 modules were responding correctly. Known loads were then applied during calibration.

The calibration error may be calculated using the equation:

$$Error(\%) = \frac{|Measured - Reference|}{Reference} \times 100\%$$

Eq. 1.11

Table 1.7: Load cell calibration test results and relative error.

<i>Reference Load (%)</i>	<i>Left Reading (kg)</i>	<i>Left Error (%)</i>	<i>Right Reading (kg)</i>	<i>Right Error (%)</i>
<i>20.0</i>	19.8	1.00	20.2	1.00
<i>40.0</i>	39.6	1.00	40.3	0.75
<i>60.0</i>	59.5	0.83	60.4	0.67
<i>73.0</i>	72.4	0.82	73.5	0.68

The final system demonstrated that both left and right channels were capable of responding independently to changes in applied load.

1.3.1.2 Left/Right Weight Distribution Test

The platform was subsequently tested using different standing conditions. Three principal conditions were evaluated:

- Approximately centered;
- Leaning towards the left;
- Leaning towards the right.

Under the centered condition, the left and right percentages remained relatively similar and ASI will decrease. However, when the test subject tends to lean towards one side, the corresponding load will increase as well as the ASI.

Table 1.8: Result obtained for weight distribution test.

<i>Condition</i>	<i>Left %</i>	<i>Right %</i>	<i>ASI (%)</i>	<i>Interpretation</i>
<i>Centered</i>	52	48	7.2	Balanced
<i>Lean left</i>	64	36	58	Left asymmetry
<i>Lean right</i>	37	63	51.2	Right asymmetry

Based on the results obtained in [Table 1.8](#), this confirms that the ASI responds logically to bilateral loading differences.

1.3.1.3 ASI Session Test

A complete session was then recorded. During the session, the webpage continuously calculated:

- Current ASI;
- Average ASI;
- Maximum ASI;
- Balanced time; and
- Session performance.

The test demonstrated that the system could summarize an entire rehabilitation session rather than purely relying on instantaneous measurements.

1.3.1.4 Audio Feedback Test

The DFPlayer audio prompts were tested under different loading conditions. The correct audio response was observed when the balance state changed beyond the defined threshold. Start, stop, and corrective audio prompts were also successfully integrated into the final system.

1.3.1.5 Web Communication Test

The /data, /start, /stop, and /tare functions were all tested. The terminal did show the presence of the command, which represented that the ESP32 received the command from the webpage. Besides that, the browser also successfully received the ESP32 data and also update the interface in real-time. This proved that the web communication is happening and is also successful. More importantly, the session state and timer were also operated correctly after the final modifications.

1.3.1.6 Firebase Storage Test

Since the completed sessions need to be saved, hence Firebase was introduced into the system. And the completed rehabilitation sessions were all saved to the Firebase Realtime Database, which preserved all the session data.

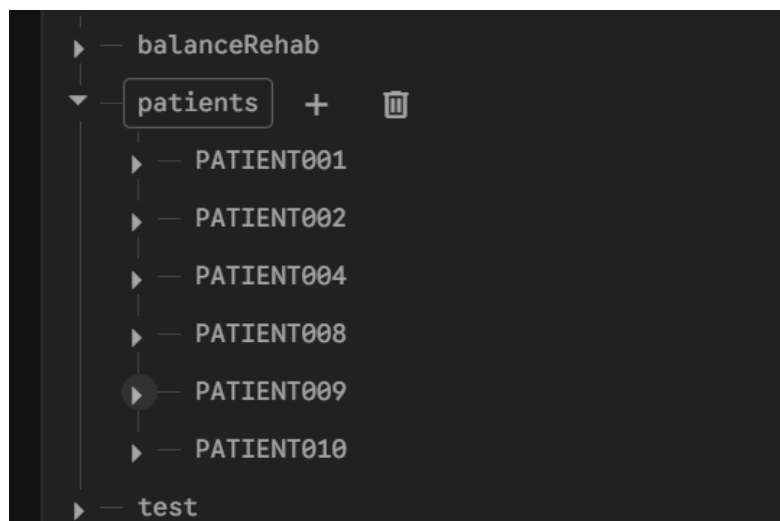


Figure 1.19: All the patient data are stored and saved to Firebase Realtime Database.

The webpage was subsequently refreshed and the previously stored data and patient information remained available. Also, when accessed into the Firebase backend, the patient data and information were all stored in the desired order and all the session data were preserved. This confirmed that the database provided persistent storage independent of the current ESP32 sessions.

1.3.1.7 Historical Data and Trend Test

To verify whether this feature is working, there are at least three rehabilitation sessions were recorded for one patient (PATIENT009). And then, the data were successfully retrieved and displayed through the historical interface. The system also generated the graphical comparison and trend assessment for that particular patient.

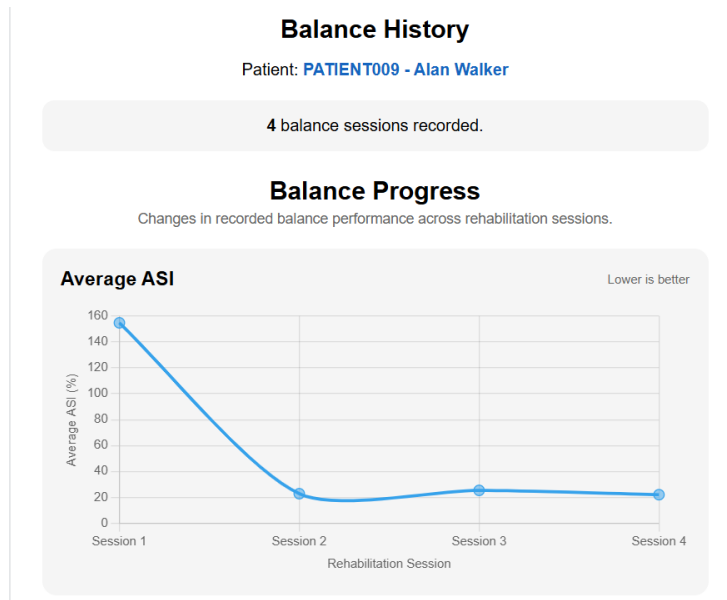


Figure 1.20: Detailed graphical analysis based on previous session history for PATIENT009.

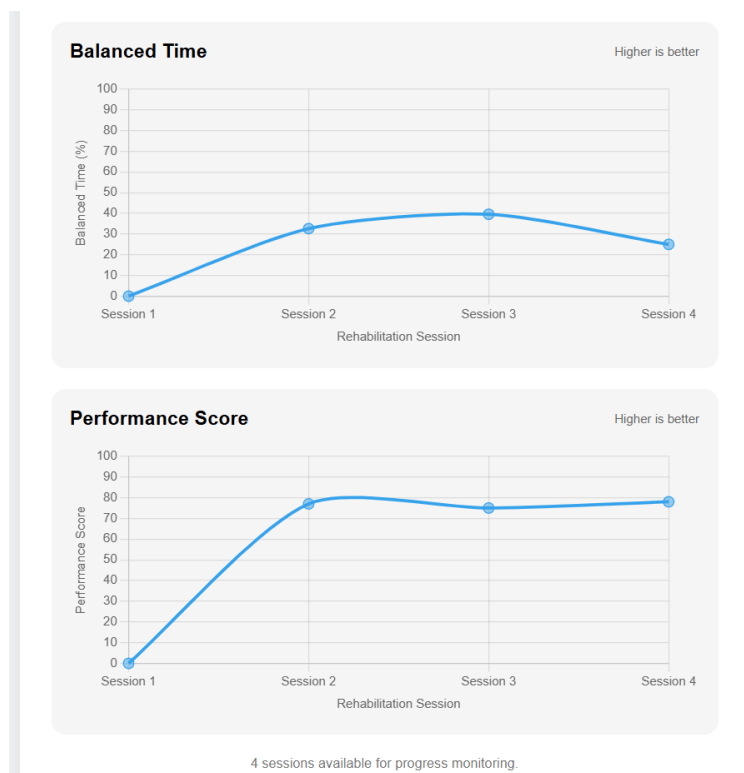


Figure 1.21: Balance time and performance score graphs based on PATIENT009's historical data.

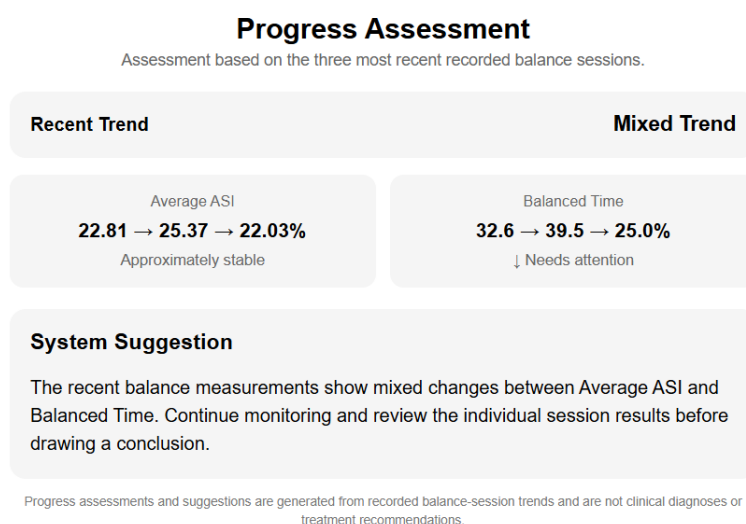


Figure 1.22: Trend assessment generated by the system based on the patient's past performance.

Session 4	
8 Sep 2026 12:52:29	
Duration	43 sec
Average ASI	22.03 %
Maximum ASI	66.04 %
Balanced Time	25 %
Performance	78 / 100
Valid Samples	40

Session 3	
4 Sep 2026 19:21:26	
Duration	48 sec
Average ASI	25.37 %
Maximum ASI	135.32 %
Balanced Time	39.53 %
Performance	75 / 100
Valid Samples	43

Session 2	
4 Sep 2026 19:04:00	

Figure 1.23: Numerical history of the patient's past rehabilitation sessions.

Therefore, the test demonstrated that the developed platform could perform longitudinal monitoring rather than only displaying real-time measurements.

1.3.2 Problems Encountered and Troubleshooting

1.3.2.1 Brownout and ESP32 Reset

The first problem that I encountered during the development phase is the unexpected ESP32 resets. The issue was found out to be associated with the unstable power delivery and the pin communication issue.

Initially, all the operations were powered directly from the PC's USB output, which might not be supplying sufficient power to maintain the ESP32 operation. To cope with this issue, a phone charger were used to supply stable 5V 2A to the ESP32 expansion board. This solution significantly reduced the reset frequency of the ESP32.

However, in later stage when more and more modules were introduced into the circuit, the problem still occur but not as frequent. After running necessary diagnostics, it was found that there is one pin from the HX711 that might be interfering with the ESP32 communication. Turns out that the pin (GPIO12) is a boot sensitive pin, and it was connected to one of the HX711 module during initial setup. This is also one of the leading causes of the ESP32 keep on resetting. And the solution is relatively simple, just by switching the pin from GPIO12 to another pin completely solved the problem.

These two solutions together maintained the system stability in the later stage, and the problem was completely solved.

1.3.2.2 HX711 Not Ready

At several stages of development, the program reported that one or both of the HX711 modules were not ready. Blocking sensor read could also slow the entire application. Therefore, the solution implemented into the system is by introducing readiness checking and timeout handling. This is to enable the system to continue servicing other operations without interruption.

1.3.2.3 Calibration Drift

This is the problem from the load cells. Initially, the load cell measurements did not consistently match the known reference weights. For example, when a 5 kg load is placed onto the platform, instead of consistently showing 5 kg, it will drift to ± 0.5 kg, which results in 4.5 kg or 5.5 kg for the 5 kg weight.

To solve this problem, the left and right channels were calibrated independently, whereby the left side is placed a load first, only then the same load is placed on the right. This could provide a better reference value to calculate the calibration factor. On top of that, a tare function was

also incorporated into the system so that the zero offset could be corrected before each rehabilitation session. And this problem is significantly minimized.

1.3.2.4 DFPlayer Initialization Failure

Initially, the DFPlayer repeatedly produced initialization failures. To dive deeper into the problem, the wiring and *UART* configuration were therefore being investigated. After the problem was studied, it turns out that stable *UART* communication can be achieved by adding an appropriate resistor in line with the communication circuit.

This added resistor had a value of $1\text{ k}\Omega$, and this significantly help the DFPlayer to communicate and function better with the ESP32, which subsequently eliminated the initialization failure issue.

1.3.2.5 Slow Web Response

In earlier version during development of prototype, the system sometimes required several seconds to respond, which seemed very “laggy” in the user’s eye. And this issue will definitely affect the user experience (UX), and it had to be fixed.

After running detailed diagnosis, it turns out that this issue happened due to the ESP32 was performing too many tasks, which seriously affecting its memory. The tasks include:

Therefore, to cope with this slow web response, the software was optimized by reducing unnecessary OLED updates, avoiding blocking sensor reads, and moving higher-level operations into the browser. As a result, all the heavy work such as metric calculations was done on the browser, whereas the ESP32 remained all the sensor acquisition and actuator control, while only reading small HTTP request from the network.

This approach significantly increased the web response time and also eliminated this issue completely.

1.3.2.6 Direct Firebase Integration Instability

In earlier architecture, it attempted to perform frequent Firebase operations directly from the ESP32. This had significantly increased network and memory demand from the ESP32 itself, which is quite burdening. When the ESP32 itself has too much load and burden, symptoms such as instability, reset behaviour, and memory-related problems started to appear.

To eliminate this problem, the architecture was then reconsidered and changed. Hence, the final architecture was as follows:



Figure 1.24: Efficient architecture for Firebase integration.

for most higher-level data operations.

The ESP32 remained primarily responsible for hardware functions and also real-time measurements. All the other heavier work was distributed to the browser, with much more efficient CPU. This structure significantly improved the reliability, stability, and usability of the system.

1.3.2.7 Session State and Timer Error

During early testing, when one session was completed, the Start/End button and timer did not always reset correctly, and they looked unprepared for the next session. Then, during diagnostic procedures, a small bug was found in the JavaScript control logic. This was a relatively simple fix, whereby just need to perform some minor changes in the JavaScript stage-management logic. Once the section is revised, the problem was fixed, and the consecutive rehabilitation sessions could be started and stopped correctly, with a fully functioning timer.

1.3.2.8 Summary of Troubleshooting

Table 1.9: Summary of problems encountered and their solutions.

<i>Problem</i>	<i>Corrective Action</i>	<i>Result</i>
<i>Brownout/reset</i>	External 5V 2A supply and switched boot sensitive GPIO	Stable ESP32
<i>HX711 not ready</i>	Readiness/timeout checks	Improved responsiveness
<i>Calibration drift</i>	Independent recalibration and tare	Improved measurement
<i>DFPlayer initialization failure</i>	UART and wiring correction	Audio operational
<i>Slow website response</i>	Reduced embedded workload	Faster response

<i>Firestore instability</i>	Browser-centered architecture	Firestore	Improved reliability
<i>Session state error</i>	Revised JavaScript logic		Correct repeated sessions

1.3.3 Limitations

Despite the developed Balance Rehabilitation Monitoring and Control Subsystem successfully achieved the intended prototype functions, several limitations remain and should be considered when interpreting the system performance.

The first limitation is that the current platform mainly evaluates left-right weight-distribution symmetry. This is suitable for the intended bilateral balance-training objective because the system is designed to identify whether the patient is leaning towards one side. However, the present configuration does not measure detailed front-back sway or generate a complete two-dimensional center-of-pressure (COP) trajectory. Therefore, the system is more suitable for monitoring bilateral loading symmetry, rather than for advanced posturographic assessment.

Next, the second limitation is related to load-cell calibration and mechanical installation. The accuracy of the left and right weight measurements depends on correct calibration, stable sensor mounting, and a rigid platform structure. Changes in the physical arrangement, sensor mounting, or zero offset may influence the load-cell readings. As a result, the system should be tared before use and recalibrated when necessary to maintain reliable measurements.

Furthermore, the third limitation is regarding the 10% ASI threshold used to define the balanced region. In the current prototype, this threshold provides a practical and consistent boundary for distinguishing relatively symmetrical and asymmetrical loading conditions. However, different patients may require different acceptable symmetry ranges depending on their condition and rehabilitation stage. Therefore, the current threshold should be considered a prototype design criterion rather than a universal rehabilitation standard.

Moreover, the fourth limitation is that the web-based monitoring and Firebase functions depend on network availability. The ESP32 can continue to perform local sensing and feedback, but remote monitoring, online session storage, historical-data retrieval, and trend assessment may be temporarily unavailable if the Wi-Fi or internet connection is interrupted.

In addition, the fifth limitation is related to the multi-session trend assessment. The system is able to compare repeated rehabilitation sessions using parameters such as average ASI, maximum ASI, balanced-time percentage, and performance score. This provides a useful indication of whether the measured balance performance is improving, remaining stable, or declining. However, the current trend assessment is based only on the metrics generated by the prototype and has not yet been compared against established clinical outcome measures.

Another limitation is that the performance score and ASI-based indicators are project-specific monitoring metrics. They are effective for comparing the user's performance within the developed system, but they are not intended to replace established clinical assessment tools. The system should therefore be used as a rehabilitation monitoring aid rather than as a diagnostic device.

The current system also relies on consistent user positioning on the platform. Significant differences in foot placement between sessions may influence the measured left-right loading distribution. Although the system can still identify asymmetry, more consistent positioning would improve the reliability of long-term comparisons. Therefore, the therapist still has to be physically present in the scene to guide the patient to step on the platform correctly to avoid obtaining misleading data.

Lastly, the developed prototype was produced for academic engineering evaluation and has not undergone any formal medical-device certification or large-scale clinical testing. Therefore, the system should be regarded as a prototype rehabilitation monitoring platform that supports supervised training rather than a replacement for professional clinical assessment.

1.3.4 Future Improvements

There are several improvements that can be done to the subsystem in future development to further enhance the sensing capability, flexibility, and practical application of the system.

One improvement that would be made is to introduce patient-specific ASI thresholds. Instead of using a fixed 10% threshold for every patient, the therapist could configure the acceptable symmetry range according to the patient's rehabilitation condition and progress. For instance, a larger tolerance can be used during early rehabilitation stage, while the target could gradually become more demanding as the patient is improving. This would make the balance feedback more personalized.

Secondly, another improvement would be upgrading the load-cell arrangement from the current left-right configuration to a four-zone measurement system. Each corner of the platform could then be measured independently. This would allow the system to evaluate both left-right and front-back loading, which makes it possible to estimate a two-dimensional center-of-pressure position (*Aryan et al., 2024*). The current ASI-based bilateral assessment could still be retained while being complemented by more detailed balance information.

Furthermore, another improvement would be made in the web application, whereby it can be enhanced by allowing the therapist to define individual rehabilitation targets. These targets could include the desired ASI range, minimum balanced-time percentage, required session duration, number of sessions, and also performance goals. The webpage could then automatically compare the recorded results against the predefined patient-specific objectives.

Next, another improvement can also be done by strengthening the historical analysis and trend-assessment functions. Instead of only comparing a small number of sessions, future versions could calculate moving averages (MA), percentage improvement from baseline, session-to-session variation, and longer-term progress trends. This would allow the therapist to obtain a more detailed overview of the patient's rehabilitation progress.

Moreover, the Firebase implementation could also be improved by introducing secure user authentication and role-based access. For example, therapists, patients, and administrators could be given different access permissions. This would make the system more suitable for handling multiple users while improving data security and privacy.

Another important improvement that would be made to this subsystem is to implement offline data storage and automatic synchronization. If the internet connection becomes unavailable, the web application could temporarily store the completed session data locally and only upload it to Firebase once the connection is restored. This would reduce the risk of losing rehabilitation data due to temporary network disruption.

On the other hand, the audio feedback system could be also improved so that it can be more flexible by allowing the therapist to select or customize patient-specific audio prompts. Features such as multilingual, feedback frequency, and instructions could be configured according to the patient's needs. This would significantly improve the accessibility and make the system more suitable for a wide range of patients.

In addition, the mechanical design of the platform could also be improved for more realistic rehabilitation use. Possible enhancements include a stronger frame, anti-slip surface, protective enclosure for the electronics, and optional handrails. These changes would definitely improve the durability and safety during repeated use.

Last but not least, testing can also be introduced to the subsystem to validate its standard and performance across industrial competitors. It can be done by testing the subsystem with a larger number of patients and comparing the results with recognized rehabilitation or force-platform measurement. By doing so, it would help to determine the repeatability and reliability of the ASI-based monitoring method and could support the development of more clinically meaningful thresholds and performance indicators.

In short, the future improvements should build on the existing strengths of the prototype rather than replacing them. The current system already provides bilateral weight measurement, ASI-based assessment, real-time web monitoring, Firebase session storage, historical comparison, and trend analysis. Future work and improvements would mainly increase personalization, measurement depth, reliability, and clinical relevance.

1.4 Conclusion

The Balance Rehabilitation Monitoring and Control Subsystem was successfully developed as a microcontroller-based rehabilitation application capable of measuring bilateral weight distribution, providing real-time feedback, recording rehabilitation sessions and supporting longitudinal progress evaluation.

Four half-bridge load cells and two HX711 modules were used to independently acquire left- and right-side loading information. An ESP32 performs the embedded sensing, communication and hardware-control functions while a 0.96-inch OLED and DFPlayer Mini provide local visual and auditory feedback.

The Absolute Symmetry Index was implemented as the principal balance-performance metric. ASI allows raw bilateral loading measurements to be converted into a normalised indication of symmetry. The system subsequently uses ASI to determine average asymmetry, maximum asymmetry, balanced-time percentage and overall session performance.

The browser-based interface considerably extends the functionality of the embedded hardware. It provides live weight-distribution monitoring, ASI display, session controls, timing,

completed-session summaries, patient records, historical graphs and rehabilitation trend assessment.

Firebase Realtime Database provides persistent online storage, allowing completed sessions to remain available for later retrieval and comparison.

A major engineering improvement arose from the decision to divide processing responsibility between the ESP32, browser and Firebase. Early attempts to perform excessive online processing directly on the ESP32 resulted in instability and delayed responses. The final distributed architecture allows the ESP32 to concentrate on real-time sensing and hardware control while the browser handles higher-level analysis and database interaction.

Several technical problems were identified and resolved during development, including ESP32 brownout, HX711 communication, load-cell calibration, DFPlayer communication, boot-sensitive GPIO selection, delayed server responses, Firebase-related instability, and session-state errors.

The completed subsystem therefore achieved the primary individual objectives of sensing, processing, feedback, online monitoring, session storage, and rehabilitation progress analysis.

Although professional balance systems remain superior for detailed posturography and clinically validated assessment, the developed prototype demonstrates the feasibility of integrating relatively low-cost sensing hardware, embedded control, web technologies and cloud-based rehabilitation records into a single connected balance rehabilitation platform.

1.5 Reference

1. Alhasan, H., Hood, V., & Mainwaring, F. (2017). The effect of visual biofeedback on balance in elderly population: a systematic review. *Clinical Interventions in Aging*, Volume 12, 487–497. <https://doi.org/10.2147/cia.s127023>
2. Aryan, R., Inness, E., Patterson, K. K., Mochizuki, G., & Mansfield, A. (2023). Reliability of force plate-based measures of standing balance in the sub-acute stage of post-stroke recovery. *Heliyon*, 9(10), e21046–e21046. <https://doi.org/10.1016/j.heliyon.2023.e21046>
3. Aryan, R., Patterson, K. K., Inness, E. L., Mochizuki, G., & Mansfield, A. (2024). Concurrent validity and discriminative ability of force plate measures of balance during the sub-acute stage of stroke recovery. *Gait & Posture*. <https://doi.org/10.1016/j.gaitpost.2024.12.001>
4. De Nunzio, A. M., Zucchella, C., Spicciato, F., Tortola, P., Vecchione, C., Pierelli, F., & Bartolo, M. (2014). Biofeedback rehabilitation of posture and weightbearing distribution in stroke: a center of foot pressure analysis. In *Functional neurology* (Vol. 29, Issue 2, pp. 127–134). PubMed. <https://pubmed.ncbi.nlm.nih.gov/25306123/>
5. Espressif Documentation. (2025). Espressif.Com. https://documentation.espressif.com/esp32_datasheet_en.pdf
6. Kim, H., Kim, H., & Shin, W.-S. (2022). Effects of Vibrotactile Biofeedback Providing Real-Time Pressure Information on Static Balance Ability and Weight Distribution Symmetry Index in Patients with Chronic Stroke. *Brain Sciences*, 12(3), 358. <https://doi.org/10.3390/brainsci12030358>
7. Ma, C., Wong, D., Lam, W., Wan, A., & Lee, W. (2016). Balance Improvement Effects of Biofeedback Systems with State-of-the-Art Wearable Sensors: A Systematic Review. *Sensors*, 16(4), 434. <https://doi.org/10.3390/s16040434>